Distribution-Aware Interactive Attention Network and Large-Scale Cloud Recognition Benchmark on FY-4A Satellite Image

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***Abstract*— Accurate cloud recognition and warning are crucial for various applications, including in-flight support, weather forecasting, and climate research. However, recent deep learning algorithms have predominantly focused on detecting cloud regions in satellite imagery, with insufficient attention to the specificity required for accurate cloud recognition. This limitation inspired us to develop the novel FY-4A-Himawari-8 (FYH) dataset, which includes nine distinct cloud categories and uses precise domain adaptation methods to align 70 419 image-label pairs (including 110 000 train/5500 test 100 × 100 size images) in terms of projection, temporal resolution, and spatial resolution, thereby facilitating the training of supervised deep learning networks. Given the complexity and diversity of cloud formations, we have thoroughly analyzed the challenges inherent to cloud recognition tasks, examining the intricate characteristics and distribution of the data. To effectively address these challenges, we designed a distribution-aware interactive-attention network (DIAnet), which preserves pixel-level details through a high-resolution branch and a parallel multiresolution cross-branch. We also integrated a distribution-aware loss (DAL) to mitigate the imbalance across cloud categories. An interactive attention module (IAM) further enhances the robustness of feature extraction combined with spatial and channel information. Empirical evaluations on the FYH dataset demonstrate that our method outperforms other cloud recognition networks, achieving superior performance in terms of mean intersection over union (mIoU). The code for implementing DIAnet is available at https://github.com/icey-zhang/DIAnet.**

***Index Terms*— Cloud recognition, distribution-aware loss (DAL), FY-4A, Himawari-8 (H08), interactive attention.**

I. Introduction

CLOUD recognition and early warning systems are integral to aviation support, contemporary meteorological forecasting, climate science, and economic infrastructure [1]. Meteorological satellite imagery, with its expansive observational scope and rapid detection capabilities, has facilitated access to vast, uninterrupted atmospheric data. This is instrumental for monitoring meteorological shifts and projecting climatic patterns. The process of cloud recognition as a pixel-level segmentation is divided into two primary tasks: the initial task is to detect the presence and ascertain the positions of clouds, then the subsequent task involves the discernment and classification of the cloud types to identify their specific categories. Pixel-level segmentation algorithms [2], [3], [4], bolstered by the advancement of deep learning technologies, have made rapid progress in the field of natural image processing. They are capable of dividing images into fine-grained pixel-level regions, thereby identifying and classifying various objects and textures within the images.

However, the recognition of clouds in satellite imagery has not yet received sufficient attention and in-depth research from the academic community, largely due to the challenges in dataset development and the complexity of designing targeted algorithms. Compared to natural images, acquiring a large volume of high-quality, annotated satellite image datasets is more difficult and costly. Satellite images often require expert knowledge for accurate annotation, especially for cloud recognition, which necessitates meteorological expertise. In addition, satellite images typically have higher spatial resolution, and broader coverage, and are more affected by atmospheric conditions, all of which add extra difficulty to cloud recognition. The identification of clouds in satellite imagery not only demands algorithms that can process large-size and high-resolution images but also requires the ability to differentiate clouds from ground objects with similar spectral characteristics, such as snow and deserts.

In recent years, extensive research has been conducted on cloud detection across various satellite platforms, including Landsat [7], the Moderate Resolution Imaging Spectroradiometer (MODIS) [8], and Sentinel [9]. Cutting-edge deep learning algorithms [10], [11], [12] have been thoroughly investigated for their efficacy in detecting clouds in satellite imagery. These algorithms demonstrate excellent capabilities in segmenting cloud boundaries and effectively detecting clouds, but they overlook the differentiation of cloud types [13] as shown in Fig. 1(a). The World Meteorological Organization (WMO) has established more fine-grained cloud classification criteria, categorizing clouds into ten genera

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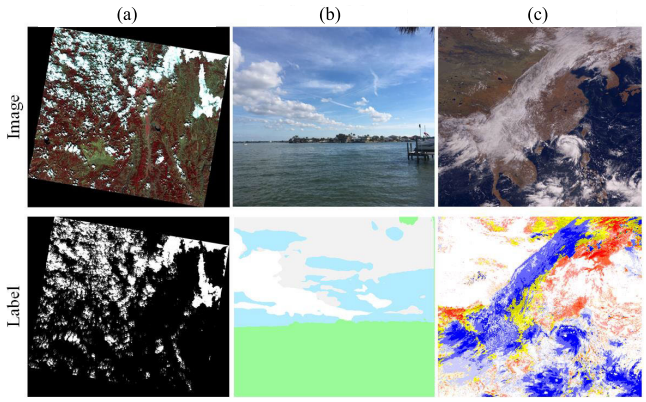


Fig. 1. Comparison of different datasets: (a) GF-1-WHU dataset [5] for satellite cloud detection which only detects the cloud field, (b) SkyCloud dataset [6] for ground-based cloud detection which only distinguishes white—thick cloud and gray—thin cloud, and (c) our proposed FYH dataset for satellite cloud recognition for nine categories specifically designed for cloud recognition tasks.

and 28 species [14]. Distinguishing between these complex cloud formations presents a much more challenge than mere detection.

To date, many algorithms [15], [16], [17] have focused on processing ground-based cloud imagery, which simplifies the task of cloud shape recognition by categorizing specific cloud types or classifying based on thickness as shown in Fig. 1(b). The coverage of ground-based imagery is limited compared to the extensive reach of satellite imagery. However, the lack of precise category labels for clouds in satellite imagery presents a significant obstacle to developing systematic algorithms for cloud classification recognition in these images. The Himawari-8 (H08) cloud-type product, operated by the Japan Meteorological Agency (JMA), has made significant strides by developing and releasing publicly available cloud shape recognition products [1]. It serves as a reliable source of expert knowledge that can be used as a valuable truth label in deep learning frameworks. Unfortunately, the cloud recognition products from H08 do not match the resolution of other satellite data, which poses challenges in the creation of datasets. Hence, constructing an effective cloud recognition benchmark and designing a viable and efficient recognition strategy is an urgent hot issue that needs to be addressed.

In this article, we introduce a benchmark dataset specifically designed for cloud recognition tasks, named FYH as shown in Fig. 1(c). The FYH dataset incorporates FY-4A L1 Data as input and utilizes H08 cloud-type products for labeling. Employing a domain adaptive (DA) operation that accounts for projection, temporal resolution, and spatial resolution, we ensure precise alignment of the data. We delve into the intrinsic challenges of cloud recognition from satellite images by conducting a thorough analysis of cloud categories and the distribution of sample categories. Our novel high-resolution keeping branch is engineered to preserve high-resolution details throughout the network, which is particularly beneficial for retaining information about small-scale features that may span as little as a single pixel. Furthermore, our parallel multiresolution cross branches are designed to extract more compact features across different resolutions and scales. This is achieved by cross-updating and fusing different branches to accurately depict the cloud outlines, capturing the inherent variability and diversity in cloud shapes. To contend with the uneven distribution across different categories, we employ a distribution-aware loss (DAL) that biases the network training toward more challenging samples. In addition, we incorporate an interactive attention module (IAM) to bolster the robustness of feature extraction, thereby enhancing the overall performance of cloud recognition.

The main contributions of this work are threefold as follows.

1) *Benchmark Production:* We believe this to be the first effort in constructing an FYH benchmark dataset with 70 419 aligned image data and labels (including 110 000 train/5500 test 100 × 100 size images). The dataset covers nine cloud categories specifically designed for cloud recognition tasks, filling a significant gap in the field.
2) *Problem Analysis:* Our work is the first to refine cloud recognition problems based on a detailed analysis of the characteristics and distributions of data, considering the complexity and variety of cloud formations.
3) *Algorithm Optimization:* We introduce a novel end-to-end network, DIAnet, which effectively accumulates local and global context dependencies across different receptive fields. Our experimental results demonstrate that DIAnet outperforms other models in this domain.

II. Related Work

This section provides an overview of significant advancements in cloud detection for satellite and ground-based images.

A. Cloud Detection for Satellite Images

Given the robust feature extraction capabilities of deep learning, numerous deep neural networks have been developed for cloud detection, resulting in significant advancements that greatly enhance detection precision and broaden applications in optical remote sensing. These models fall into three main categories: 1) threshold-based approaches; 2) machine learning-based techniques; and 3) deep learning-based methods [23]. Among these, the earliest cloud detection models are threshold-based methods [24], [25], which differentiate clouds from clear-sky regions using predefined thresholds based on the distinct high reflectivity and low-temperature attributes of clouds. For instance, Sun et al. [26] introduced a universal dynamic threshold cloud detection algorithm applicable to MODIS and Landsat-8 images, while Zhai et al. [27] proposed a unified cloud/shadow detection model for multispectral/hyperspectral images based on spectral indices and spatial matching. Despite their widespread use due to simplicity, these methods face challenges in adapting to complex scenarios where determining suitable thresholds is difficult, thereby limiting their efficacy. On the other hand, machine learning techniques identify clouds by learning significant differences in spectral and spatial features between clouds and background [5]. For example, Li et al. [5] combined threshold segmentation with multifeature guided filtering to effectively

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detect clouds and shadows in GaoFen-1 (GF-1) imagery. Utilizing support vector machine (SVM) classifiers [28], [29] for cloud/noncloud discrimination has yielded promising outcomes. In addition, methods like clustering analysis [30], Bayesian models [31], and random forests [32] have been employed in cloud detection. Nevertheless, these shallow models relying on handcrafted features often struggle to capture the intrinsic cloud characteristics and lack robustness to significant changes, leading to limited detection accuracy.

In recent years, deep learning has showcased remarkable prowess in feature extraction, heralding a new era in cloud detection [33]. By harnessing robust high-level semantic features through hierarchical representations, deep learning methodologies have spearheaded advancements in cloud detection techniques. For instance, Chai et al. [34] employed a deep convolutional neural network (CNN) to craft the AdaptSegNet model tailored specifically for Landsat-8 cloud/shadow detection. Sun et al. [35] amalgamated minimum component transformation with CNN, resulting in enhanced accuracy. Jeppesen et al. [36] introduced the RS-Net within the U-Net framework, showcasing outstanding performance for Landsat-8 imagery. Moreover, Zhan et al. [37] utilized a fully CNN (FCN) to distinguish clouds and snow, yielding precise outcomes for GF-1 imagery. Further enhancing cloud detection efficiency for GF-1 imagery, Zhang et al. [10] integrated wavelet transform-based up and down blocks with multiscale spatial attention. Another noteworthy contribution by Li et al. [38] introduced a weakly supervised cloud detection network adept at handling high-resolution imagery like GF-1 and Ziyuan-3 (ZY-3) images. Addressing challenges in cloud boundaries and perplexing areas, Zhao et al. [39] integrated a boundary prediction branch. RD-UNet [40] has been proposed for cloud detection in remote sensing images, leveraging cascaded U-shaped networks and enhanced skip connections to improve information flow, optimize edge details, and efficiently utilize multiscale features. In addition, synthetic aperture radar (SAR) images that are not affected by clouds are used as auxiliary data and paired with multicloud optical images to form a generative adversarial network (GAN) generator to directly reconstruct the cloud-free image [41]. Furthermore, an increasing number of cloud detection networks continue to emerge, dedicated to enhancing identification accuracy and efficiency. These algorithms demonstrate excellent capability in segmenting cloud boundaries and effectively detecting clouds, but they overlook the differentiation of more complex cloud types.

B. Ground-Based Cloud Recognition

While sky segmentation is frequently used as an auxiliary resource in different applications, it was rarely a primary research focus. The introduction of the Skyfinder dataset by Mihail et al. [42] composed of 100 000 images from 53 stationary webcams changed that and provided a reference point for further developments. The authors used the dataset to compare the results of three existing sky segmentation approaches. The results showed a general need for improvements and highlighted the influence of lighting and weather conditions on segmentation accuracy. Building upon this dataset, La Place et al. [43] used RefineNet [44] to improve segmentation accuracy. To increase flexibility, Nice et al. [45] presented an adaptive approach that automatically selects a suitable candidate from 13 conventional image processing techniques (e.g., Sobel filters and shift segmentation), based on the input image. However, the reported overall accuracy is still relatively low, limiting its use in real-life scenarios. Compared to sky classification, which merely distinguishes between sky and nonsky areas, cloud segmentation further subdivides the sky region into clear sky and cloudy segments. Conventional methods for cloud segmentation typically extract color features from images containing only sky and then categorize cloud and sky regions based on predetermined thresholds [46], [47]. However, these methods are constrained to specific conditions and lack the adaptability to accommodate diverse environmental settings and cloud types. To enhance adaptability, Liu et al. [48] devised a method wherein the image is partitioned into irregular segments, and a local threshold is computed for each segment. Similarly, Li et al. [49] employed a hybrid approach utilizing minimum cross-entropy and fixed thresholds. Gerhardt et al. [6] proposed SkyCloudNet, a multitask neural network architecture that extracts high-level attributes from the input image and utilizes them to improve the robustness of the network to environmental influences. Although these algorithms simplify cloud shape recognition tasks by classifying specific cloud types or based on thickness, compared to the extensive coverage of satellite images, the coverage of ground images is limited, which restricts their application in scenarios requiring large-scale observation areas, such as climate environment and meteorological monitoring.

III. Benchmarks and Methodology

A. Benchmarks

In recent years, we have witnessed the rapid evaluation of cloud detection benchmarks including the SPARCS [18], GF-1-WHU dataset [5], 38-Cloud [19], 95-Cloud [50], KappaZeta [20], CloudSEN12 [21], and CHLandsat8 [22]. Table I lists the profiles of these datasets, including satellite, scene, scale, categories, channels, resolution, and size. The SPARCS dataset [18] was originally derived manually from precollection Landsat-8 scenes. Its purpose is to validate cloud and cloud shadow masking derived from the spatial procedures for automated removal of cloud and shadow (SPARCS) algorithm. The GF-1-WHU dataset [5] is the most commonly used publicly available dataset for cloud detection, contributed by Chinese GF-1 wide field of view (WFV) imagery. The 38-Cloud dataset [19] contains 38 Landsat-8 scene images and has been cropped into 8400 patches for training and 9201 patches for testing. The 95-Cloud dataset [50], an extension of the 38-Cloud, includes 57 Landsat-8 scenes, mostly located in North America, for training purposes. The KappaZeta dataset [20] is designed for cloud and cloud shadow detection with 14 multispectral channels covering the Northern European terrestrial area at a 10-m resolution. The CHLandsat-8 dataset [22] is a high-resolution cloud detection dataset,

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TABLE I
ILLUSTRATION OF FYH CLOUD RECOGNITION DATASET AND EXISTING SATELLITE CLOUD DETECTION DATASETS AND GROUND-BASE CLOUD DETECTION

Category A: 0:clear 1:cloud 2:cloud shadow 3:shadow over water 4:snow/ice 5:water 6:land 7:flooded 8:thin cloud 9:thick cloud
Category B: 0:Clear, 1:Ci, 2:Cs, 3:Dc, 4:Ac, 5:As, 6:Ns, 7:Cu, 8:Sc, 9:St, 10:Unknown

Dataset	Year	Satellite	Scene	Image number	Category	Channel	Resolution	Image Size	Crop Size
SPARCS [18]	2014	Landsat-8	80	-	A:0,1,2,3,4,5,6,7	10	-	8000×8000	1000×1000
GF1-WHU [5]	2017	GaoFen-1	108	-	A:0,1,2	4	16m	-	-
38-Cloud [19]	2019	Landsat-8	38	8,400Train/9,201Test	A:0,1	4	-	8000×8000	384×384
95-Cloud [19]	2020	Landsat-8	95	3,4701Train/9,201Test	A:0,1	4	-	8000×8000	384×384
KappaZeta [20]	2021	Sentinel-2	155	-	A:0,1,2,8	14	-	-	512×512
CloudSEN12 [21]	2022	Sentinel-1,2	-	49,400	A:0,2,8,9	13	10m	-	512×512
CHLandsat8 [22]	2023	Landsat-8	64	22,616Train/10,080Test	A: 0,1	3	-	8000×8000	352×352
SkyCloud [6]	2023	-	-	350	A: 0,8,9	3	-	-	-
FYH	2023	FY-4A,H08	70,419	110,000Train/5,500Test	B	14	4km	1092×2748	100×100

comprising 64 comprehensive scenes captured by Landsat-8 satellites across various regions of China, and includes manually annotated cloud masks. These datasets primarily focus on cloud detection and differentiate between thin and thick clouds based on cloud density. However, the lack of categories for different cloud morphologies hinders the broader application of these datasets to more complex, higher level recognition tasks involving various cloud types.

Advanced Geostationary Radiation Imager (AGRI) is one of the main loads of the FY-4A stationary meteorological satellite. It has two independent scanning mirrors for performing north-south and east-west scans. The orbital center of FY-4A/AGRI is located at 104.7°E. The system can achieve minute-level regional fast scanning, and obtain 14 band Earth cloud images with a 15-min time interval full-disk observation mode. The bands, which cover the range of wavelengths from 0.45 to 13.8, are sampled at a nominal 0.5/1 km spatial resolution at the nadir in the visible (VIS), 2 km in the near-infrared (NIR), and 4 km in the infrared (IR) spectral bands. L1 SDR Data[1] of FY-4A/AGRI that resolution is 4 km, describes the preprocessed product that after quality control, Earth navigation, and calibration from level 0 raw package data. The level 1 data can be directly applied in the NWP model and satellite data assimilation research, the climate model study, also be widely used in the atmospheric vertical sounding (such as the vertical profiles of temperature and water vapor), and cloud detection, the atmospheric component and omit longwave radiance exploration and so on.

H08 is a new geostationary meteorological satellite operated by JMA. Its observation data[2] was released freely on July 7, 2015. The orbital center of H08/AHI is located at 140.7°E and is focused on the main areas of Japan and the Pacific Ocean (from 80°E to 160°W and from 60°N to 60°S). Its observation frequencies are every 10 min for the full disk and every 2.5 min for the area adjacent to Japan. The resolutions of the 16 bands (three visible, three NIR, and ten IR) are 0.5, 1, and 2 km, respectively. H08/AHI has more infrared bands with a higher spatial resolution than does FY-4A/AGRI, in particular, a visible band at 0.51 for true color image synthesis. Table II summarizes the information of the multispectral images between FY-4A and H08 meteorological satellites.

TABLE II
INFORMATION OF THE MULTISPECTRAL IMAGES BETWEEN FY-4A AND H08 METEOROLOGICAL SATELLITE

FY-4A/AGRI			H08/AHI			
No	Band (μm)	Spatial resolution (km)	No	Band (μm)	Spatial resolution (km)	Primary application
1	0.47	1.0	1	0.47	1.0	Aerosol
/	/	/	2	0.51	1.0	Aerosol
2	0.65	0.5	3	0.64	0.5	True color
3	0.86	1.0	4	0.86	1.0	Fog, cloud
4	1.37	2.0	/	/	/	Vegetation
5	1.60	2.0	5	1.61	2.0	Cirrus
6	2.25	2.0	6	2.25	2.0	Cloud, snow
7	3.8(high)	2.0	7	3.88	2.0	Aerosol
8	3.8(low)	4.0	/	/	/	Fire, land, and surface
9	6.29	4.0	8	6.24	2.0	Fire, land, and surface
/	/	/	9	6.94	2.0	Water vapor
10	7.12	4.0	10	7.35	2.0	Water vapor
11	8.62	4.0	11	8.60	2.0	Water vapor
/	/	/	12	9.63	2.0	Water vapor, cloud
/	/	/	13	10.4	2.0	Ozone
12	11.0	4.0	14	11.24	2.0	Cloud
13	12.09	4.0	15	12.38	2.0	SST, cloud
14	13.55	4.0	16	13.28	2.0	SST, cloud

"Himawari L1 Gridded data (NetCDF4 format)" as well as the "Himawari Standard Data (HSD format)" provided by the JMA. Himawari L1 Gridded data is generated by JAXA/EORC from the Himawari Standard Data with resampling to equal

[1]http://satellite.nsmc.org.cn/portalsite/Data/Satellite.aspx
[2]http://www.jma-net.go.jp/msc/en/

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TABLE III
INFORMATION OF THE MULTISPECTRAL IMAGES BETWEEN FY-4A L1 DATA AND H08 CLOUD-TYPE PRODUCT

	FY-4A L1 Data	H08 Cloud-type Product
File Type	HDF	NetCDF
Projection	NOM	EQR
Observation Area	China proper	60°S-60°N,80°E-160°W
Temporal Resolution	15min	10min
Spatial Resolution	4km (Pixel number:1092 Line number:2748)	5km (Pixel number:2401, Line number:2401)

latitude-longitude grids. The Geophysical Parameter Data estimated from the geostationary H08 Standard Data and Model parameters are available in near-real-time. Available Geophysical Parameters include Aerosol Property, Sea Surface Temperature, Cloud Property, WildFire, Nighttime Sea Surface Temperature, Chlorophyll-a, and Short-Wave Radiation/Photosynthetically Available Radiation. The cloud types defined by the International Satellite Cloud Climatology Project (ISCCP) from cloud property are utilized as labeled masks for model training and testing. Each pixel in a cloud-type image is labeled as one of 11 classes with values ranging from 0 to 10 representing: 0—clear, 1—cirrus (Ci), 2—cirrostratus (Cs), 3—deep convection (Dc), 4—altocumulus (Ac), 5—altostratus (As), 6—nimbostratus (Ns), 7—cumulus (Cu), 8—stratocumulus (Sc), 9—stratus (St), and 10—unknown.

B. Domain Adaptive

For cloud recognition for FY-4A satellite imagery, the cloud-type products from H08 are employed as the benchmark for performing validations. We produce a DA on projection, temporal resolution, and spatial resolution for the match consistency of the FY-4A L1 Data and H08 cloud-type product as shown in Table III. To the best of our knowledge, this is the first time that such match consistency datasets have been used for the validation of a new development cloud recognition method. To ensure the aligned consistency between FY-4A/AGRI L1 Data and H08/AHI cloud-type production, we applied domain adaptation techniques on projection, temporal resolution, and spatial resolution. One of the challenges we encountered was the difference in projection between the FY-4A and H08 satellites. To address this, we transformed the FY-4A NOM projection to the EQR projection, with a spatial resolution of 0.05°. The FY-4 satellite adopts the geostationary orbit nominal projection defined by the CGMS LRIT/HRIT global specification, and its geographic coordinates are calculated based on the WGS84 reference ellipsoid. The Algorithm 1 clarifies the process of projection transformation.[3]

This transformation, as shown in Fig. 2(b) and (d), enabled us to achieve consistency in the spatial representation of the data. To ensure a consistent input format for our model, we unified the common areas between 85° E to 134.5° E and 5° N to 49.5° N with size of 900 × 1000, as illustrated in Fig. 3(a) and (c). These areas were selected as the input region for our model. The data from the year 2020 was chosen as the research objective. In addition, due to the unavailability of H08 cloud-type products at night in the specified area, we only utilized the data from 00:30 UTC to 10:50 UTC in a day. To ensure temporal consistency, we controlled the time difference between the FY-4A and H08 data within 2 min. For

[3]https://fy4.nsmc.org.cn/nsmc/cn/satellite/FY4A.html

Algorithm 1 Transformation of the FY-4A Nominal (NOM) Projection to the Equal-Rectangular (EQR) Projection

Input: Geographic longitude lon and latitude lat; $\pi = 3.141592653589793238 4626$; The semi-major axis of the Earth $ea = 6378.137[km]$; The minor axis of the Earth $eb = 6356.7523[km]$; The distance from the center of the Earth to the satellite center $h = 42164[km]$; The longitude of the satellite subpoint $\lambda_D = 104.7$; Column offset COFF; Column scale factor CFAC; Line offset COFF; Line scale factor CFAC;

Output: The line number l and column number c.

1: The angular representation of geographical longitude and latitude needs to be converted into radians.

$$\text{lon} = \frac{\text{lon} \times \pi}{180}, \quad \text{lat} = \frac{\text{lat} \times \pi}{180} \tag{1}$$

2: The longitude and latitude coordinates are transformed into geocentric coordinates.

$$\lambda_e = \text{lon}, \quad \phi_e = \arctan\left(\frac{eb^2}{ea^2} \times \tan(\text{ lat })\right) \tag{2}$$

3: Calculate the r_e.

$$r_e = \frac{eb}{\sqrt{1 - \frac{ea^2 - eb^2}{ea^2} \times \cos^2(\phi_e)}} \tag{3}$$

4: Calculate the r_1, r_2, r_3.

$$\begin{aligned} r_1 &= h - r_e \times \cos(\phi_e) \times \cos\left(\lambda_e - \lambda_D\right) \\ r_2 &= -r_e \times \cos(\phi_e) \times \sin\left(\lambda_e - \lambda_D\right) \\ r_3 &= r_e \times \sin(\phi_e) \end{aligned} \tag{4}$$

5: Calculate the r_n, x, y.

$$\begin{aligned} r_n &= \sqrt{r_1^2 + r_2^2 + r_3^2} \\ x &= \arctan\left(-\frac{r_2}{r_1}\right) \times \frac{180}{\pi} \\ y &= \arcsin\left(-\frac{r_3}{r_n}\right) \times \frac{180}{\pi} \end{aligned} \tag{5}$$

6: Calculate the c, l.

$$\begin{aligned} c &= COFF + x \times 2^{-16} \times CFAC \\ l &= LOFF + y \times 2^{-16} \times LFAC \end{aligned} \tag{6}$$

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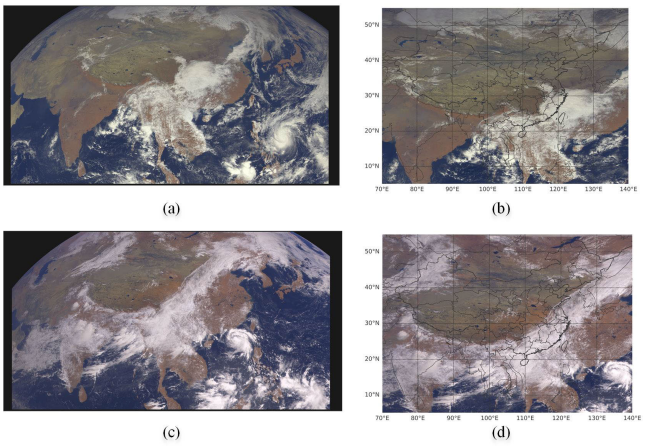


Fig. 2. Projection transformation: (a) and (c) is NOM projection and (b) and (d) is the result of projection transformation to EQR projection.

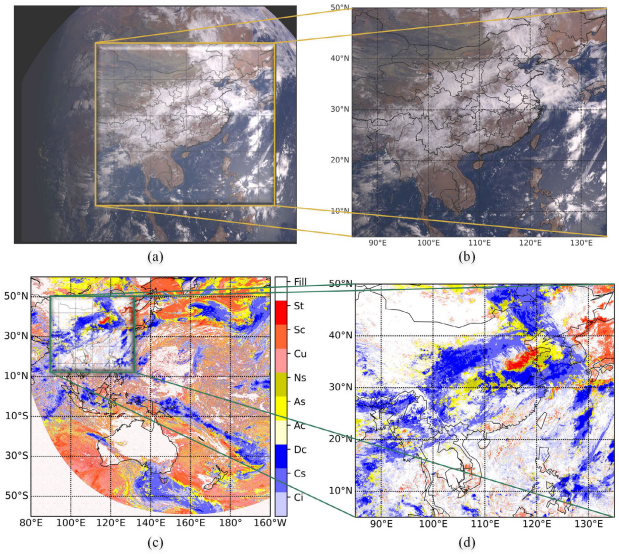


Fig. 3. DA: (a) FY-4A L1 Data, (c) H08 cloud-type product, and (b) and (d) result of DA. The class unknown and clear are merged into class fill for display.

example, the H08 cloud-type product data at 1:20 on July 18, 2020, shown in Fig. 3(c), was matched with the FY-4A L1 data displayed in Fig. 3(a), where the observation started at 1:19 on July 18. The result of the domain adaptation is shown in Fig. 4.

The FY-4A AGRI L1 Data underwent several preprocessing steps, including quality control, Earth navigation, and calibration from the level 0 raw package data. To normalize the range of pixel values in the images, we applied maximum normalization without radiation calibration, resulting in values normalized to the range of [0, 1]. The normalization process is represented as follows:

$$N_i^t = \left(I_i^t - \mathrm{Min}\left(I_i^t\right)\right) / \left(\mathrm{Max}\left(I_i^t\right) - \mathrm{Min}\left(I_i^t\right)\right) \tag{7}$$

where N_i^t denotes the normalized image of the ith band at time t, I_i^t represents the adaptive image of FY-4A L1 data, and $\mathrm{Min}(I_i^t)$ and $\mathrm{Max}(I_i^t)$ are the minimum and maximum pixel values in the ith band at time t, respectively. Each band, ranging from band01 to band05 representing albedo, and band06 to band14 representing brightness temperature, was normalized individually.

By implementing these preprocessing techniques, we guarantee the consistency and normalization of the input data, thereby facilitating accurate and dependable cloud-type classification using the FY-4A/AGRI L1 Data. Following screening, projection, and cropping procedures, the 70 419 data points from the year 2020 are partitioned into a total of 110 000 training images and 5500 validation images, each sized at 100 × 100 pixels. The FYH dataset is illustrated in Fig. 5.

C. Distribution of Sample Categories

Cloud recognition poses a challenging task as it requires distinguishing clouds with diverse shapes and types at the pixel level. The complexity of clouds further adds to the difficulty of this problem. To gain insights into the data properties and provide a practical and interpretable solution, we conducted a data distribution analysis, as depicted in Fig. 6.

Through the analysis, we identified several key issues that affect cloud recognition performance.

Issue (1): First, as shown in Fig. 6(a), some cloud objects are small and may even occupy just a single pixel. When the resolution decreases, these small objects risk being lost or overlooked, leading to inaccurate recognition results.

Issue (2): Second, the distribution of clouds is random, and their shapes are not fixed. The inherent variability and diversity of cloud shapes make it challenging to accurately delineate cloud outlines. The ability to handle varying cloud shapes is crucial for achieving precise cloud recognition.

Issue (3): Finally, as illustrated in Fig. 6(b), the distribution of cloud categories is imbalanced within the dataset. Class 0, which represents clear skies, contains a large amount of data, resulting in a "long tail distribution" for the entire dataset. This imbalanced distribution can cause issues during the learning process, as the model tends to be biased toward training on noncloud data, which is not desired.

D. Distribution-Aware Interactive-Attention Network

We have focused on designing an end-to-end data-driven network for cloud recognition to address the issues discussed above in the dataset. Consequently, we introduced a method called DIAnet. Fig. 7 illustrates a general overview of the DIAnet framework. Intuitively, the proposed DIAnet jointly trains branches in different resolutions in an end-to-end pattern.

1) Baseline: We introduce HRNet [51] as the baseline of the DIAnet as depicted in Fig. 7, to maintain a high-resolution representation throughout the network, which significantly aids in effectively addressing the issue (1) by preserving pixel-level details. Note that, We abandon the stem operation that consists of two strides 3 × 3 convolutions at the beginning because this will cause the resolution to be reduced to 1/4, resulting in the loss of pixel targets. Some improvements designed on baseline include an IAM and DAL.

Most of the existing fusion schemes primarily consist of low-level and high-level fusion methods. However, DIAnet

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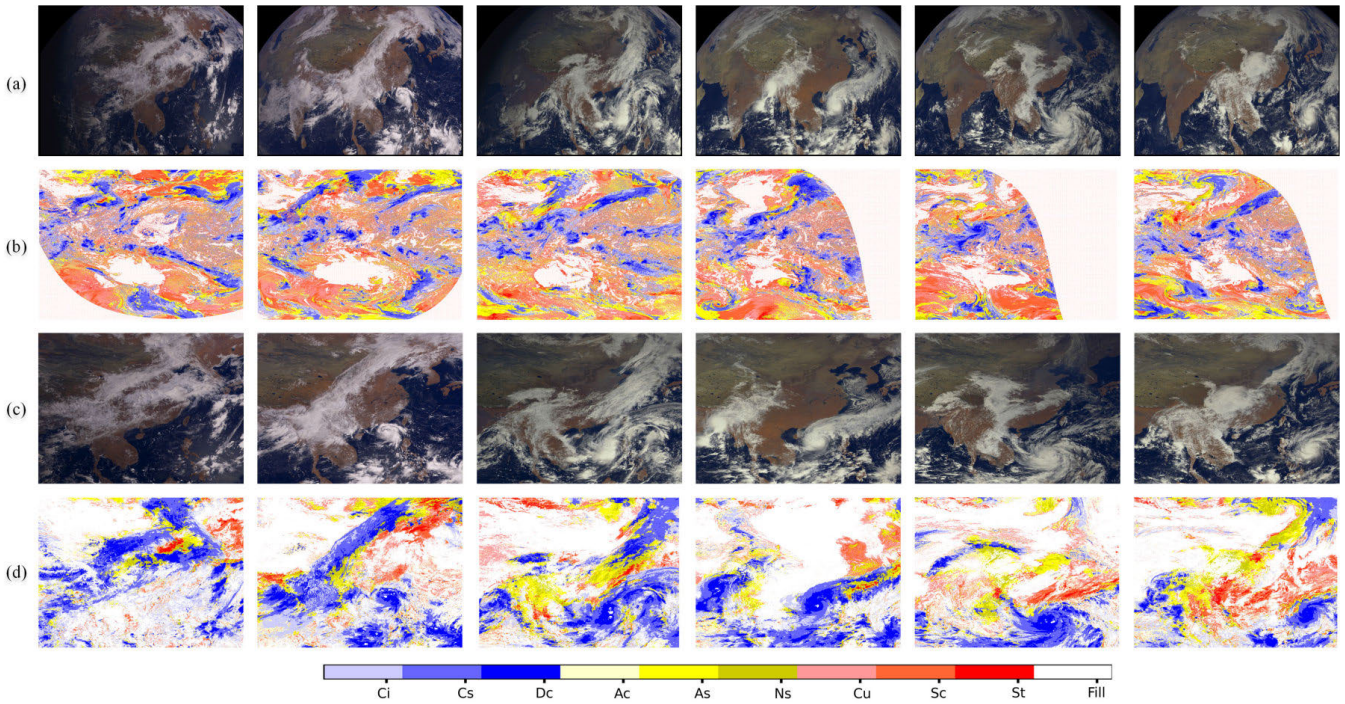


Fig. 4. Result of the DA: (a) FY-4A L1 Data, (b) H08 cloud-type product, and (c) and (d) result of DA. The class unknown and clear are merged into class fill for display.

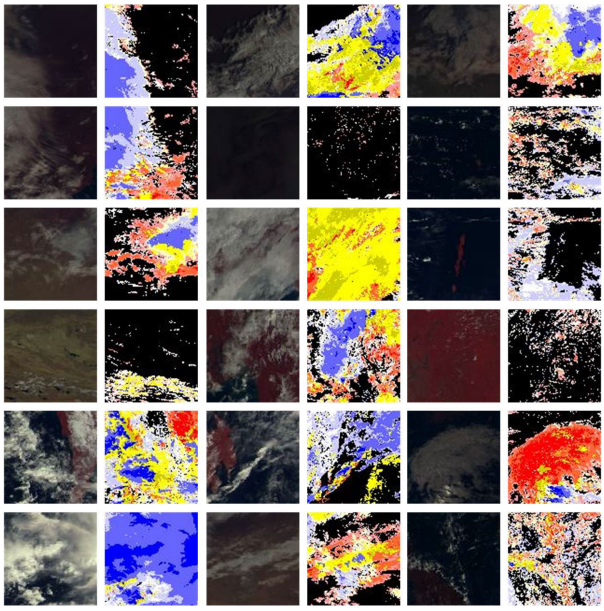


Fig. 5. FYH dataset.

takes a different approach from the commonly used feature fusion architectures. It incorporates multiscale fusion repeatedly, leveraging low-resolution (high-resolution) representations of the same depth and similar level to enhance the high-resolution (low-resolution) representation. This approach enriches the high-resolution representation, making it more suitable for cloud recognition. In addition, the parallel multiresolution cross branches in DIAnet aim to learn more compact features of different resolutions and scales by cross-updating the fusion of different branches. This configuration allows the network to not only learn specific properties from its own resolution but also consider diverse supplements from another stream, leading to a more comprehensive blending of information. Furthermore, the high-level features are converted into low-level features through the process of upsampling. This enables the network to effectively transfer higher level information to lower level representations.

The idea behind the four branches is to use features extracted at different scales, providing both local and global context. We arranged the four stages because early layers' feature maps retain more spatial details, resulting in refined boundaries, while deeper layers extract high-level semantic features. This allows us to obtain features from different receptive fields to overcome the issue (2).

2) Interactive Attention Module: In addition, the IAM, as shown in Fig. 8, plays a crucial role in cloud recognition tasks. It can capture correlations between different spatial positions and channels and effectively fuse them. In addition, by introducing the interactive attention mechanism, we can better model the shape, texture, and context of clouds, thereby providing more comprehensive spatial-channel information.

Specifically, to reduce computational costs while modeling relationships within the contextual region, we devised a spatial attention module, as depicted in the blue box in Fig. 4. Given a feature input $X \in \mathbb{R}^{C\times H\times W}$, we partition it into four subfeature maps $X_k \in \mathbb{R}^{C\times H/4\times W/4}$ in the spatial dimension (where $k = 1, 2, 3, 4$). Subsequently, these maps are fed into two 1-D convolutions with a channel compression ratio of r, resulting in A_k and D_k for each subbranch. The upper branch undergoes covariance matrix operations to yield a spatial attention map, which is then applied as weights on the lower branch. The formula is as follows:

$$\mathcal{F}_k = \text{softmax}\left(A_k \bar{I} A_k^T\right) * D_k \tag{8}$$

where $\bar{\mathbf{I}} = (1/n)(\mathbf{I} - (1/n)\mathbf{1})$, $n = H/4 \times W/4$, $\bar{\mathbf{I}}$ and $\mathbf{1}$

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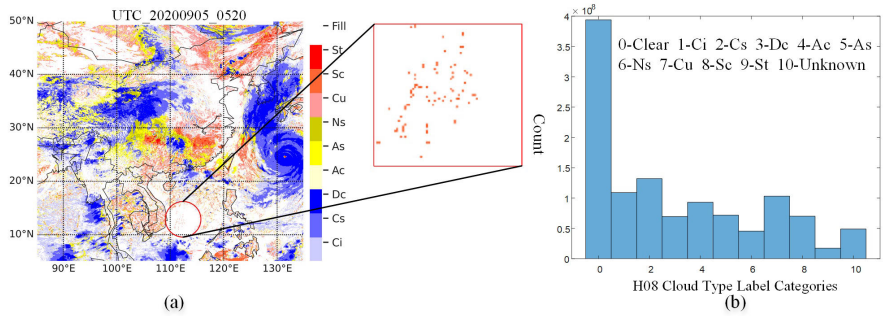


Fig. 6. Illustration of cloud categories and the distribution of sample categories. (a) H08 cloud-type product. (b) Cloud categories distribution.

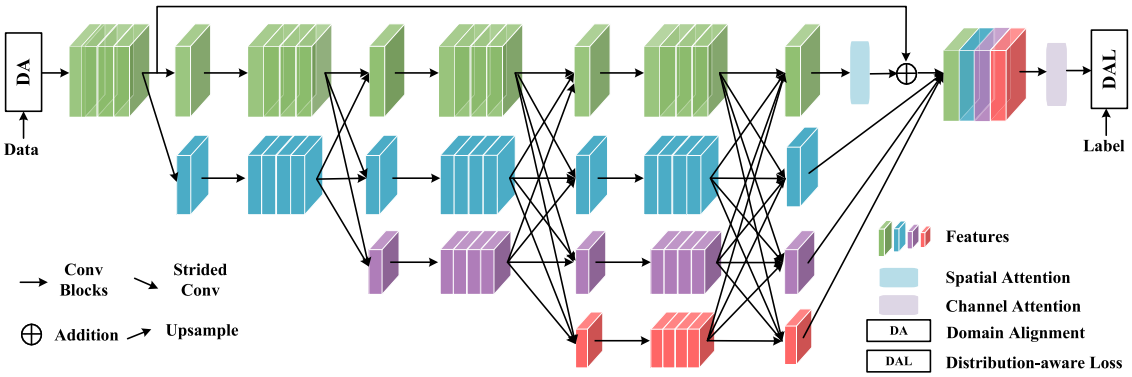


Fig. 7. Illustration of the proposed end-to-end DIAnet. It incorporates multiscale fusion repeatedly, leveraging low-resolution (high-resolution) representations of the same depth and similar level to enhance the high-resolution (low-resolution) representation. In addition, the parallel multiresolution cross branches in DIAnet aim to learn more compact features of different resolutions and scales by cross-updating the fusion of different branches. DAL automatically selects hard examples and strengthens the training of these samples in unbalanced cloud category datasets.

represent the $n \times n$ identity matrix and matrix of all ones. Finally, after channel-wise restoration, a residual connection is utilized to enhance spatial information, yielding

$$\mathcal{Z}_k = \text{conv}(\mathcal{F}_k) + X_k \tag{9}$$

after refining all subfeature maps X_k, we obtain an overall feature map that encapsulates rich contextual information.

In addition, to model the spatial information in the channel dimension and obtain more powerful feature correlation learning, a channel attention module is proposed. It selectively emphasizes informative features by exploring adaptive-weighted feature statistics. As illustrated in the purple box in Fig. 4, given a set of intermediate feature maps denoted as $X = [x_1, x_2, \ldots, x_C]$, containing C feature maps of size $H \times W$, X is concatenated in the channel dimension to obtain a size of $\mathbb{R}^{C \times H \times W}$. Then, a single-channel 1-D convolution layer is utilized to learn an adaptive-weighted matrix, followed by a softmax layer for normalization before multiplication with the input. The formula is as follows:

$$Y = \text{softmax}(\text{conv}(X)) * X \tag{10}$$

where $Y = [y_1, y_2, \ldots, y_C]$ represents the channel descriptors. To more fully utilize the aggregated information after weighting and feeding it back into the input features, we reduce its channel dimension by a factor of t, then scale it back to the original dimension, inserting an activation function in between. This is known as a simple gating mechanism with a sigmoid function. The obtained channel attention map is then applied to the input features, and the calculation process is given by the following equation:

$$V = \text{sigmoid}(\text{conv}_c(\text{PRelu}(\text{conv}_{c/t}(Y)))) * X. \tag{11}$$

$V = [v_1, v_2, \ldots, v_C]$ is the output feature map of the channel attention module, $\text{conv}_{c/t}$ and conv_c denote 1-D convolutions with output channels of c/t and c, respectively. It collaborates with the spatial attention module to embed spatial information into the channels. It adaptively adjusts channel features for recalibration, promoting representation learning in the network.

3) Distribution-Aware Loss: To mitigate the influence of issue (3), we introduced a simple yet remarkably effective OHEM algorithm [52] as our DAL. The procedure for DAL involves several key steps. First, for a given batch of N samples, we compute the loss $\mathcal{L}(V_i, y_i)$ for each sample i, where V_i represents the input and y_i denotes the corresponding label. This is achieved using the cross-entropy loss function

$$\mathcal{L}(V_i, y_i) = -\log p_{y_i}(V_i) \tag{12}$$

where $p_{y_i}(V_i)$ is the predicted probability of the true class y_i for sample V_i. Next, the samples are ranked in descending order based on their loss values, and the top K samples with the highest losses are selected, where $K = \lfloor \text{ratio} \times N \rfloor$ and ratio is a predefined fraction (between 0 and 1) that specifies the proportion of hard examples to be mined. The DAL loss,

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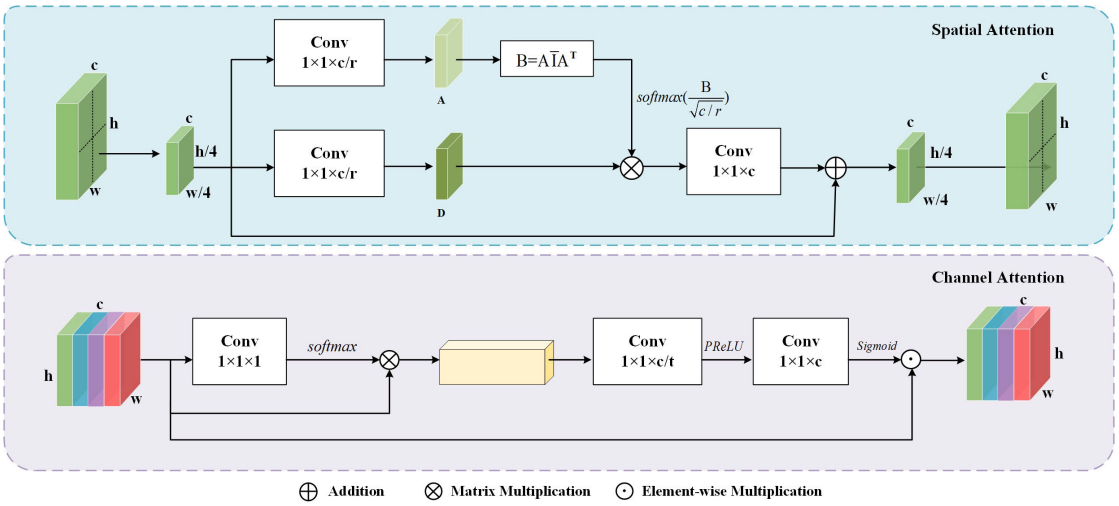


Fig. 8. Structure of the IAM. It can capture correlations between different spatial positions and channels and effectively fuse them to model the shape, texture, and context of clouds, thereby providing more comprehensive spatial-channel information.

TABLE IV
COMPARISON OF RECOGNITION IOU RESULTS (%) WITH DIFFERENT BACKBONE ON THE FYH DATASET

Backbone	mIoU	Clear	Ci	Cs	Dc	Ac	As	Ns	Cu	Sc	St	Unknown
Resnet50 [53]	35.33	62.01	27.90	48.73	43.18	16.15	37.29	34.46	32.66	47.21	23.21	15.83
ViT [54]	36.64	58.91	40.35	58.66	60.42	19.07	33.54	39.94	20.10	30.86	35.04	6.09
HRNet [51]	38.18	65.58	32.08	53.84	48.38	16.22	34.50	32.74	36.11	**48.52**	22.05	18.26
HRNet without stem	**45.21**	**68.38**	**47.74**	**66.58**	**66.33**	**27.28**	**42.85**	**47.59**	**31.23**	35.48	**38.82**	**25.09**

TABLE V
EFFECTS OF EACH COMPONENT IN OUR DIANET WITH MIOU RESULTS (%). CA REPRESENTS CHANNEL ATTENTION AND SA REPRESENTS SPATIAL ATTENTION

Baseline	DAL	IAM(SA)	IAM(CA)	mIoU
✓				45.21
✓	✓			47.73
✓		✓		46.38
✓			✓	46.20
✓		✓	✓	47.00
✓	✓	✓	✓	**49.86**

$\mathcal{L}_{\text{DAL}}$, is then calculated as the mean loss of these selected hard examples

$$\mathcal{L}_{\text{DAL}} = \frac{1}{K}\sum_{j=1}^{K}\mathcal{L}(V_{i_j}, y_{i_j}). \tag{13}$$

This algorithm automatically selects hard examples and strengthens the training of these samples in unbalanced datasets, leading to improved learning efficiency and robustness.

IV. EXPERIMENT

A. Experiment Setting

1) Dataset: We generate our FYH dataset with FY-4A L1 Data and H08 cloud-type products as illustrated in Section III-A. After the screening, projecting, and cropping processing, the 70 419 data in the 2020 year are divided into a total of 110 000 training images and 5500 test images, with the size of 100 × 100.

2) Accuracy Metrics: The accuracy assessment measures the agreements and differences between the recognition result and the reference mask at the pixel level. The intersection over union (IoU) and mean IoU (mIoU) are used as metrics to evaluate the performance of the compared methods. The calculation of these metrics is illustrated as follows:

$$\text{IoU} = \frac{\text{TP}}{\text{TP}+\text{FP}+\text{FN}} \tag{14}$$

$$\text{mIoU} = \frac{1}{n+1}\sum_{i=0}^{n}\frac{p_{ii}}{\sum_{j=0}^{n}p_{ij}+\sum_{j=0}^{n}p_{ji}-p_{ii}} \tag{15}$$

where the true positive (TP) and true negative (TN) denote a correct prediction, and false positive (FP) and false negative (FN) denote an incorrect outcome. p_{ii} is used to represent the original class i and predict class i at the same time, that is, and TN, p_{ij} denotes that class i was originally predicted to be class j, that is, FP and FN. Note that the mIoU are comprehensive indicators, while precision and recall are related to the commission and omission errors, respectively. And the larger value means higher performance.

3) Implementation Details: Our model is implemented using the Pytorch library. We utilize the SGD optimizer to train the model with a weight decay of 0.0005 and a momentum of 0.9. The number of epochs is set to 100 with the batch size of 32. We adopt the learning rate of 0.01. The whole training process is on 4 NVIDIA GeForce RTX 2080Ti GPUs.

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TABLE VI
COMPARISON OF RECOGNITION IOU RESULTS (%) WITH DIFFERENT ATTENTION METHOD ON THE FYH DATASET

Method	mIoU	Clear	Ci	Cs	Dc	Ac	As	Ns	Cu	Sc	St	Unknown
PSP [57]	47.64	69.02	51.01	**70.46**	69.06	**30.57**	41.73	56.32	37.29	45.89	36.18	16.52
EMA [58]	46.24	68.15	49.04	67.05	67.97	29.36	45.31	51.11	30.96	43.50	38.54	17.70
OCR [59]	48.98	70.29	51.08	68.93	65.72	30.33	**48.83**	51.81	31.61	47.04	**47.47**	25.70
CBAM [60]	46.67	**69.05**	48.98	67.49	68.94	28.65	44.63	47.46	31.49	43.42	37.37	**25.89**
Ours (IAM)	**49.86**	**69.05**	**49.04**	69.57	**70.41**	30.60	43.71	**57.95**	**40.18**	**49.32**	46.41	22.21

TABLE VII
COMPARISON OF RECOGNITION IOU RESULTS (%) WITH DIFFERENT LOSS FUNCTION ON THE FYH DATASET

Loss Function	mIoU	Clear	Ci	Cs	Dc	Ac	As	Ns	Cu	Sc	St	Unknown
ACW [55]	47.96	68.22	**50.93**	**70.25**	67.87	29.99	43.61	56.83	38.43	**50.17**	38.56	12.65
Cross Entropy [56]	47.73	67.76	49.71	68.05	67.87	29.83	**45.12**	51.41	32.20	45.36	45.58	22.12
Ours (DAL)	**49.86**	**69.05**	49.04	69.57	**70.41**	**30.60**	43.71	**57.95**	**40.18**	49.32	**46.41**	**22.21**

TABLE VIII
COMPARISON OF RECOGNITION IOU RESULTS (%) WITH DIFFERENT METHODS ON THE FYH DATASET

Method	mIoU	Clear	Ci	Cs	Dc	Ac	As	Ns	Cu	Sc	St	Unknown
DeepLab v3+ [61]	35.21	60.92	38.81	56.28	56.45	17.78	31.78	35.29	21.66	32.24	33.09	3.02
Unet [62]	36.64	58.91	40.35	58.66	60.42	19.07	33.54	39.94	20.10	30.86	35.04	6.09
Unet++ [63]	37.37	68.26	36.39	60.86	59.73	21.67	34.62	29.42	23.78	35.32	27.24	13.76
Unet+HFSA [64]	39.14	61.89	42.03	61.25	62.57	21.26	36.36	42.35	24.67	34.22	31.41	12.60
Gated-SCNN [65]	41.03	62.61	44.30	62.96	64.29	24.61	38.86	44.30	26.45	34.24	34.34	14.36
Mask2Former [66]	41.55	71.39	39.86	63.07	60.01	25.28	38.50	37.58	26.79	40.58	35.13	18.90
RD-UNet [40]	43.64	**74.15**	41.31	65.25	65.18	21.97	41.89	50.81	29.52	44.03	41.37	4.59
DIANet	**49.86**	69.05	**49.04**	**69.57**	**70.41**	**30.60**	**43.71**	**57.95**	**40.18**	**49.32**	**46.41**	**22.21**

B. Ablation Studies

1) Validation of the Baseline Framework: The experiment results presented in Table IV demonstrate the performance of different backbone architectures on the FYH dataset in terms of recognition IoU. Four backbone architectures, namely, ResNet50 [53], ViT [54], HRNet [51], and HRNet without stem, were evaluated. ResNet50 achieved an mIoU of 35.33%, with varying IoU scores across different cloud types. It demonstrated relatively high IoU scores for Clear (62.01%) and Cs (48.73%) categories but lower scores for other categories. HRNet exhibited improved performance compared to ResNet50 and ViT, achieving an mIoU of 38.18. It showed notable improvements in several categories, particularly in Ci (32.08%), Dc (48.38%), and Sc (48.52%). HRNet without stem outperformed both ResNet50 and HRNet, achieving the highest mIoU of 45.21. It demonstrated superior performance across most categories, with significant improvements in clear (68.38%), Ci (47.74%), Cs (66.58%), and Dc (66.33%). Overall, HRNet without stem showed the best performance on the FYH dataset, indicating its effectiveness in cloud recognition tasks and we use it as the baseline of the DIAnet. This also validates the importance of maintaining high-resolution representations throughout the entire network with the high-resolution preservation branch. By preserving pixel-level details, this helps effectively address some very small cloud objects that may be lost or overlooked during the network's feature extraction process due to the reduction in resolution, thereby leading to inaccurate recognition results.

2) Effects of Each Component: To understand how each module in DIAnet promotes cloud recognition accuracy, we test each component independently on the baseline HRnet and report the performance in Table V. The 2.52% increase in result gain comes from DAL adaption, verifying the importance of the DAL to overcome the issue of unbalanced category distribution. When IAM is added to HRNet, performance improves by 1.79% mIoU. The final mIoU reaches 49.86% when all three steps are adopted, representing a 4.65% absolute enhancement and validation of the effectiveness of our approach.

3) Compare of Different Attention Module: Table VI presents a comparative analysis of recognition IoU results obtained using various attention methods on the FYH dataset. While existing methods like PSP [57], EMA [58], OCR [59], and CBAM [60] achieved competitive mIoU scores ranging from 46.24% to 48.98%, IAM outperformed them all with the highest mIoU of 49.86%. Combining IAM, our approach demonstrated superior performance in accurately recognizing different cloud types, highlighting its robustness and effectiveness in cloud-type recognition tasks. This indicates that the

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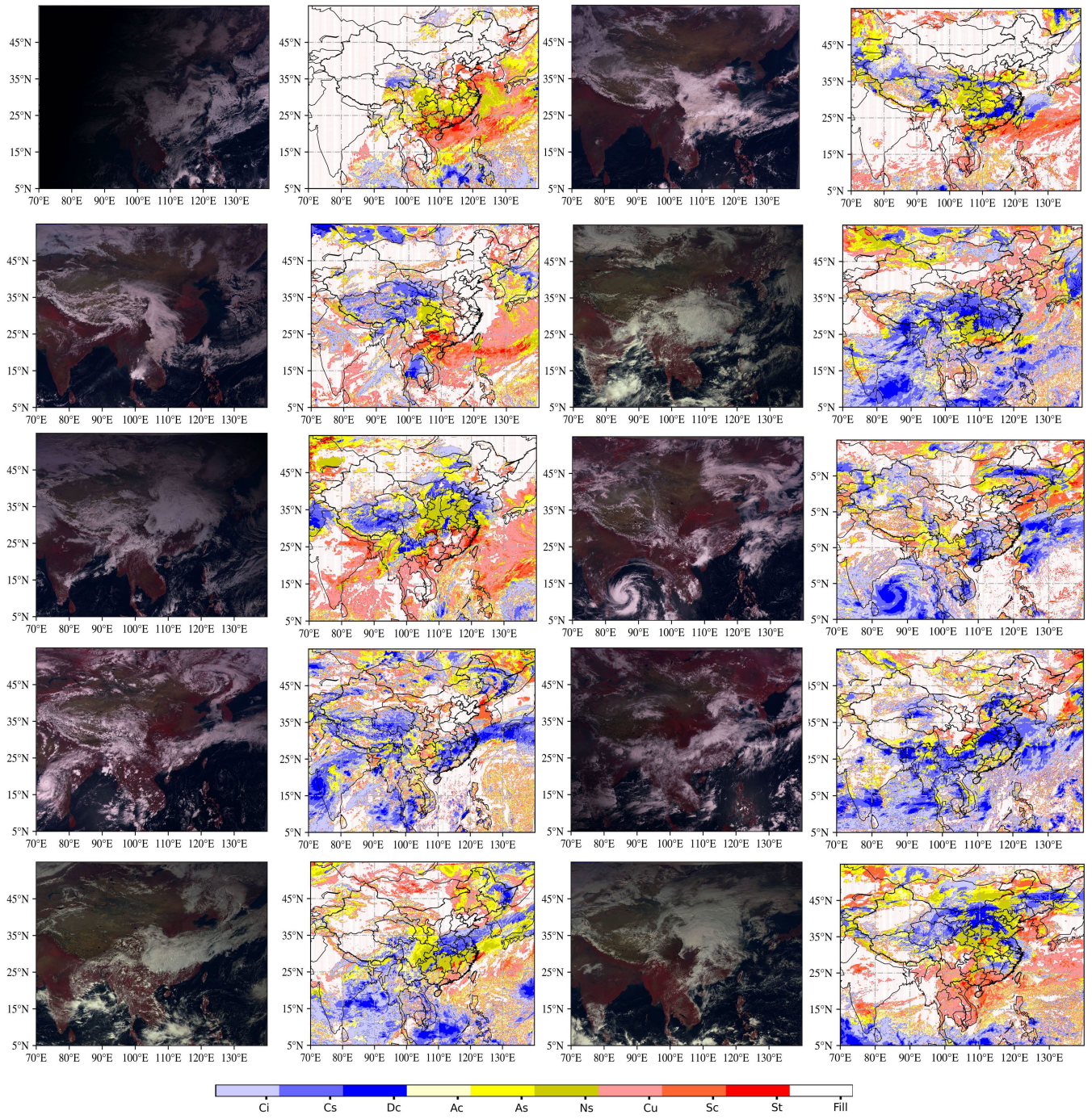


Fig. 9. Cloud-type products generated by our proposed DIAnet on the FY-4A L1 EQR projection data.

IAM plays a crucial role in cloud recognition tasks. It can capture correlations between different spatial positions and channels and effectively fuse them. In addition, by introducing the interactive attention mechanism, we can better model the shape, texture, and context of clouds, thereby providing more comprehensive spatial-channel information.

4) Compare of Different Loss Function: The effectiveness of various attention mechanisms in cloud recognition tasks was evaluated using the FYH dataset as illustrated in Table VII. Methods including ACW [55], Cross Entropy [56], and our proposed DAL were compared based on mIoU and IoU scores for different cloud types. Results indicate that our proposed DAL method achieved the highest mIoU of 49.86%, surpassing ACW and Cross Entropy. Our method also demonstrated superior performance in segmenting specific cloud types such as clear, Ci, Cs, Dc, and Sc. The incorporation of DAL enabled our method to effectively capture complex cloud structures and contextual information, leading to improved segmentation accuracy. These results highlight the effectiveness of our proposed attention mechanism in enhancing cloud recognition performance on the FYH dataset.

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TABLE IX
QUANTITATIVE EVALUATION OF COMPARISON EXPERIMENTS (%) ON THE CLOUD DETECTION DATASET

Methods	IoU	Model Size
U-Net [62]	76.48	17.3M
DeepLabv3+ [61]	75.57	40.4M
RS-Net [36]	76.43	12.3M
Cloud-Net [19]	75.83	36.5M
Boundary Net [68]	76.42	88.9M
RD-UNet [40]	79.67	31.5M
DIAnet	82.09	66.0M

C. Comparison With Other Methods

As shown in Table VIII, we compare DIAnet with other popular methods on our FYH dataset. The proposed method improves the performance by 6.21% compared to the second-best method. The performance of various segmentation methods was evaluated on the FYH dataset, considering metrics such as mIoU and IoU scores for different cloud types. DeepLab v3+ [61], Unet [62], Unet++ [63], Unet + HFSA [64], Gated-SCNN [65], Mask2Former [66], and RD-UNet [40] were among the methods compared. Results show that DIANet outperformed all other methods, achieving the highest mIoU of 49.86%. DIANet demonstrated significant improvements in recognizing various cloud types, including Ci, Cs, Dc, Ac, As, Ns, Cu, Sc, St, and unknown, showcasing its effectiveness in cloud segmentation tasks on the FYH dataset. These results highlight the superiority of DIANet in cloud recognition tasks compared to existing segmentation methods.

D. Visualization of Cloud-Type Products

We test the ten complete images on the FY-4A/AGRI L1 EQR projection data with the size of 900 × 1000 (5° N to 49.5° N, 85° E to 134.5° E) to generate the final cloud-type products. As shown in Fig. 9, our product can effectively sense cloud regions and obtain cloud category results at a pixel level with fine granularity. This demonstrates the importance of maintaining high-resolution features and extracting multiscale features in high-resolution networks.

E. Generation Experiment on the Cloud Detection Dataset

We utilized the SPARCS dataset [67] to demonstrate the superiority of DIAnet in the cloud detection task. The SPARCS dataset consists of 80 scenes of ten-band Landsat-8 data, each with a size of 1000 × 1000 pixels. The dataset was split in a 3:1 ratio for training and testing, and the images were further cropped into 200 × 200 patches, resulting in 1500 patches for training and 500 patches for testing. The number of epochs is set to 200 with the batch size of 6. We adopt the learning rate of 1e-4. The whole training process is on an NVIDIA GeForce RTX A100 GPU. Table IX presents a comparative analysis of various cloud detection models based on IoU and model size. DIAnet stands out with the highest IoU of 82.09%, indicating superior accuracy in cloud detection, despite its larger model size of 66.0M. RD-UNet follows closely with an IoU of 79.67% and a more moderate model size of 31.5M, suggesting a good balance between performance and resource requirements. The U-Net and RS-Net show comparable IoU scores, with U-Net having a slightly larger model size of 17.3M compared to RS-Net's 12.27M. DeepLabv3+, Cloud-Net, and Boundary Net have lower IoU scores but exhibit a wide range of model sizes.

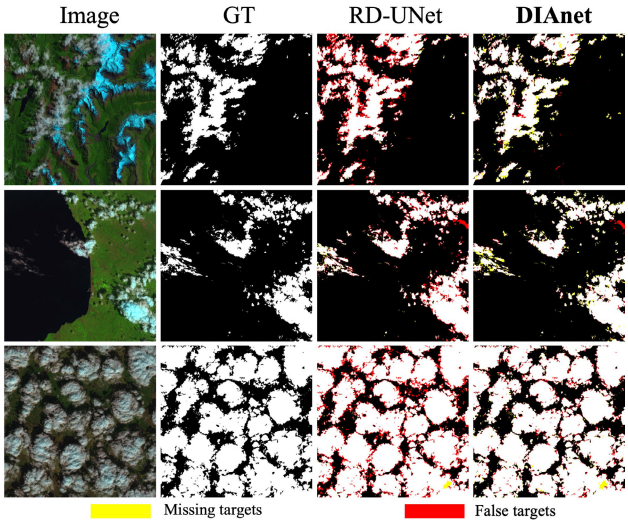


Fig. 10. Visual results of cloud detected by RD-UNet [40] and DIAnet on SPARCS dataset.

Fig. 10 illustrates that DIAnet consistently achieves cloud detection results that closely match the ground truth under various conditions. It stands out in handling complex challenges, including the identification of thin clouds and the differentiation of bright terrain features. In contrast, RD-UNet's detection results may exhibit a higher rate of false detections. Nevertheless, the computational requirements of DIAnet could be restrictive in resource-limited environments, particularly when compared with those of less complex network models. In addition, we provide the computational efficiency results for our method evaluated under the condition using an NVIDIA GeForce RTX 3090 GPU with 24 GB of VRAM and an input size of ×100 × 100 with 14 channels. The number of parameters, FLOPs, and FPS of the DIANet are 66.0M, 59.19G, and 2.95.

V. CONCLUSION

In conclusion, this article addresses the challenges in cloud recognition tasks and presents innovative solutions to overcome them. By constructing the FYH dataset, we address the scarcity of cloud category datasets in satellite cloud imagery, enabling deep learning networks to not only detect clouds but also recognize them. Our proposed DIAnet model effectively handles the difficulties posed by small cloud targets and varying cloud shapes, as well as imbalanced distributions between different cloud categories. Through the utilization of high-resolution branches, parallel multiresolution cross branches, and the incorporation of a DAL, our DIAnet model achieves outstanding performance on the FYH dataset. These

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techniques allow for accurate and robust cloud recognition, even in challenging scenarios. Furthermore, the introduced IAM enhances the robustness of feature extraction, leading to a continuous improvement in cloud recognition accuracy. This module enables the model to focus on relevant cloud features and discard irrelevant information, resulting in more precise and reliable cloud recognition. Overall, our study contributes to the field of cloud recognition by addressing dataset limitations and proposing effective strategies to improve cloud recognition accuracy. The performance of our DIAnet model on the FYH dataset demonstrates its effectiveness in handling various cloud recognition challenges. These findings have implications for satellite imagery analysis, weather forecasting, and other applications where accurate cloud recognition is essential.

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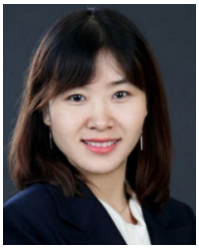
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